

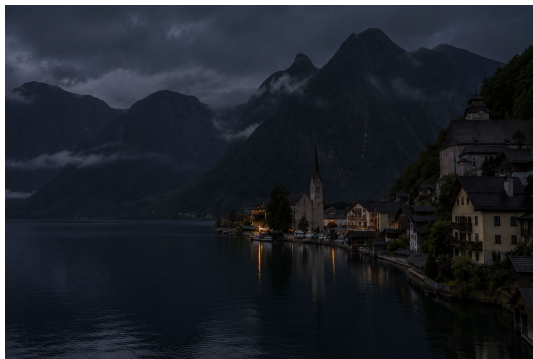
Lab2

1. Histogram Equalization (50%)

- 計算輸入圖的直方圖
- 計算直方圖的累計表
- 用直方圖累計表完成各強度的映射

Intensity (r)	No. of Pixels	Axx Sum of P	Output value	Quantized Output(s)
0	20	0.2	$0.2 * 7 = 1.4$	1
1	5	0.25	$0.25 * 7 = 1.75$	2
2	25	0.5	$0.5 * 7 = 3.5$	4
3	10	0.6	$0.6 * 7 = 4.2$	4
4	15	0.75	$0.75 * 7 = 5.25$	5
5	5	0.8	$0.8 * 7 = 5.6$	6
6	10	0.9	$0.9 * 7 = 6.3$	6
7	10	1	$1.0 * 7 = 7$	7
Total	100			

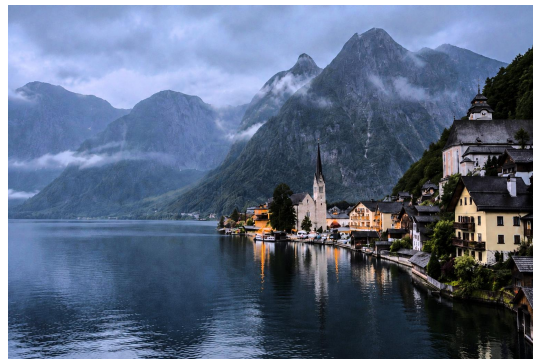
input



bad output



good output



1-a 分別對BGR三個顏色作直方圖等化 (15%)

- 顏色會跑偏

input



output



1-b 對HSV圖片的V做直方圖等化 (35%)

- hint: `cv2.cvtColor()`

output



$$h = \begin{cases} 0^\circ & \text{if } \max = \min \\ 60^\circ \times \frac{g-b}{\max-\min} + 0^\circ, & \text{if } \max = r \text{ and } g \geq b \\ 60^\circ \times \frac{g-b}{\max-\min} + 360^\circ, & \text{if } \max = r \text{ and } g < b \\ 60^\circ \times \frac{b-r}{\max-\min} + 120^\circ, & \text{if } \max = g \\ 60^\circ \times \frac{r-g}{\max-\min} + 240^\circ, & \text{if } \max = b \end{cases}$$

$$s = \begin{cases} 0, & \text{if } \max = 0 \\ \frac{\max-\min}{\max} = 1 - \frac{\min}{\max}, & \text{otherwise} \end{cases}$$

$$v = \max$$

2. Otsu Threshold (50%)

- 先計算影像的直方圖
- 把直方圖強度大於閾值的像素分成一組, 把小於閾值的像素分成另一組
- 分別計算這兩組的組內變異數, 並把兩個組內變異數相加
- 將 0 ~ 255 依序當作閾值來計算組內變異數和, 總和值最小的就是結果閾值

2. Otsu Threshold (50%)

Input



Output

